LAKSHMI NARAIN COLLEGE OF TECHNOLOGY (MCA), BHOPAL

MCA – I SEMESTER

Information Technology

MCA-104

Assignment

UNIT-I (CO-I)

Case Study 1: Smart College Campus Using Wi-Fi and Bluetooth

An LNCT-MCA college wants to develop a smart campus. Students require internet access in classrooms, laboratories, libraries and hostels. Faculty members use wireless projectors, Bluetooth keyboards, wireless headphones and smartphones. The college administration wants to connect multiple devices while maintaining reliable communication.

Question:

As a system designer, explain the complete working of this system. Your answer must cover the following points:

i. Explain how Wi-Fi can provide wireless internet connectivity across the campus.
ii. Differentiate between Wi-Fi and Bluetooth based on range, data transfer, power consumption and applications.
iii. Recommend whether Wi-Fi or Bluetooth should be used to connect a laptop to a wireless projector. Explain your choice.
iv. How can the college use Wi-Fi and Bluetooth together to improve campus services?
v. Identify two limitations of wireless communication and suggest suitable measures to address them.

OR

Case Study 2: Communication Between Two Cities Using Microwave Systems

A telecom company wants to establish a communication link between two cities separated by a mountainous region. Installing underground cables is expensive and time-consuming. The company is considering microwave communication because it can provide high-capacity links without requiring a continuous physical cable.

Questions:

As a system designer, explain the complete working of this system. Your answer must cover the following points:

i. Explain the working principle of a microwave communication system.
ii. Why is line-of-sight communication important in terrestrial microwave systems?
iii. Explain the role of transmitting antennas, receiving antennas and repeaters in the system.
iv. Compare microwave communication with radio communication and fibre-optic communication.
v. Recommend a suitable communication method for this situation, explaining its advantages and limitations.

Note:

i. Give suitable examples and diagrams wherever necessary.
ii. Make neat and clean assignment file clearly mentioning your name, roll number, section, and Subject code with subject name at the front page.